Related work for each white-paper task

ONR YIP full proposal planning, Oct 26, 2026. Start with the one-page gist; each task section follows.

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The gist

Every task now has a classical result behind it, which answers Pedja's "ad hoc" concern. The 20-page draft's propositions mostly hold but need restating. The main novelty claims survive the search: no work found picks debaters and judges impartially, has a judge re-run the debaters' tool calls, or pairs planted-case audits with random audits of real decisions.

Task	Classical result to lean on	Closest recent work	Most important change
1.1 Counting shared mistakes once	Optimal vote weights (Nitzan and Paroush; Shapley and Grofman 1984); identifiability (Kruskal; Allman et al. 2009); Chow's reject rule for when to debate	MiniRep (Sept 2026), CARE (ICML 2026), Hossain et al. 2026	State P1 in general form; P2 holds only up to relabeling, anchored by planted cases
1.2 Impartial debaters and judges	Impartial selection (Holzman and Moulin; Alon et al. 2011; Fischer and Klimm 2015, best ratio 1/2); randomized assignment caps (Jecmen et al. 2020)	PeerRank (Feb 2026), MiniRep, Lev et al. 2023	Raters must be same-answer outsiders or neutral; fix the judge pool before answers
2.1 Debate before an impartial judge	Refereed delegation of computation (Canetti et al. 2011/2013), deployed as blockchain fraud proofs; Dung's argumentation semantics	Jiang et al. 2026, Tool-MAD, DynaDebate, PROClaim	Recast the stopping rule as refereed delegation over one shared outline of steps
2.2 Audits against collusion	Inspection games (Dresher; Maschler; von Stengel); Steinhardt et al. 2016; anytime-valid tests (Ville's inequality)	AI control (Greenblatt et al. 2024; Gardner-Challis et al. 2026)	Split P5: audit coverage bound plus a detection limit that grows with the square root of the number of decisions, since honest agents also err

Task	Classical result to lean on	Closest recent work	Most important change
3 Rewards from group decisions	Peer prediction and correlated agreement (Dasgupta and Ghosh 2013; Shnayder et al. 2016; Liu and Guo 2020); potential-based shaping	Co-RL (Yang et al., Aug 2026), TTRL, Shafayat et al. 2025, Kenton et al. 2026	Do not credit both debaters for moving the judge toward the decision; it pays one side to argue badly
Evaluation	Control levels (Korbak et al. 2025); AI-control games (Griffin et al. 2024)	Schaeffer et al. 2026 (attack selection)	Red team must choose when to attack; MEQA is single-document as published

Fix in the white paper before reusing its text.

- PeerEval as "the first leaderboard on which AI models rank one another" is not supported: Language-Model-as-an-Examiner (June 2023) came first. Say PRD was "among the first".
- Kenton et al. 2026 found that debate training reduced reward hacking. Invented quotes appeared only with a deliberately weakened judge.
- Shu 2026 does not say systems "discuss until they agree"; it shows unanimity rising from 39.5% to 95.2% after one round, with accuracy nearly unchanged.
- Kurokawa et al. 2015 is not a partition mechanism, and PRD is not impartial.
- "No existing debate protocol holds up when agents share mistakes, lie, or collude" overreaches, because Byzantine-robust consensus among LLM agents already tolerates some faulty agents.
- Diyi Yang holds a 2024 ONR YIP in the same Mathematical Data Science program, on trustworthy human-AI agent interaction. Say how this project differs.

Still open. The web-search limit cut short the August-October 2026 sweep in every thread, so re-run it before October 30. The sections below are drafts; an independent check of their citations and numbers is still under way.

Task 1.1: Deciding what to debate, counting shared mistakes once

For Task 1.1, which weighs answers by track record and sends weakly supported questions to debate, the 20-page draft's cap on a correlated group's evidence, Proposition P1, holds and is tight. Its yes/no bound, $\log((1 - \lambda)/\lambda)$ for a shared hidden answer wrong with probability λ, is the classical optimal vote weight of one voter with accuracy $1 - \lambda$ (Nitzan and Paroush 1982; Shapley and Grofman 1984), so P1 makes exact, for any group size, the classical advice to count a correlated bloc as one voter. **The most important change is to restate P1 in its general form, with the equal-likelihood formula as a corollary, and P2 as identification up to relabeling, anchored by planted cases.**

Prior work and its limits

Independence fails for LLM raters.

- Dawid-Skene gives each rater a confusion matrix, the probability of each reported answer given each true answer, fitted without labels by EM, an iterative method; raters are independent given the truth (Dawid and Skene 1979). Zhang et al. 2016 (JMLR) start EM from a spectral estimate built from three groups of raters, with an error rate optimal up to a log factor; the proofs use independence.
- LLM errors are correlated, more so in larger, more accurate models (Kim et al. 2025, ICML). Nine judges were worth 2.18 independent votes on MNLI, and 9.1% of the 319 items all nine agreed on were wrong, against about 0.02% under independence (Kohli 2026, preprint, whose datasets "remain classification or binary preference tasks"). Dawid-Skene rated GPT-5.2's valence labels 90.7% accurate, against 71.0% on human gold labels (Said and Villaneau 2026, LREC 2026 workshop).
- For Task 1.1: these measurements need labels, and established aggregation closed at most 11% of Kohli's gap to independent judges even with correct answers, so the task estimates dependence without labels and seeks new evidence by debate.

Copying and same-model groups.

- Dong et al. 2009 (VLDB) detect copying among data sources from shared false values, since "sharing the same false value is typically a rare event", but warn that "sharing common data does not in itself imply copying".

- Jaffe et al. 2016 (AISTATS) place a hidden binary variable between the truth and each group of classifiers, the binary form of Task 1.1's model. They warn that independence-based accuracy estimates "may be highly inaccurate" under dependence, and call their own model "in general not identifiable": different parameter settings fit the same data.
- MiniRep (Zhang et al. 2026, preprint) weights agents in multi-agent discussion by reputation, counting each same-model group, declared in advance, once with a capped weight. It won all 28 attack conditions in a 10-agent setting, but has no judge, audits or theorem.
- For Task 1.1: Jaffe et al. is the model's binary ancestor, Dong et al. justify its "same unusual answers" signal, and MiniRep is the nearest LLM system.

Escalation in LLM systems.

- DOWN (Eo et al. 2025, preprint) starts a discussion only when one agent's confidence, its average token probability, is at or below a threshold; agreement plays no part.
- JuryProbe (Zhou and Lin 2026, preprint) estimates a panel's "consensus risk" from judges' correlated false negatives on a labeled probe, and high-risk panels re-evaluate their majority accepts against trusted references. Unanimous false consensus then fell to zero, though the authors give no formal risk guarantee.
- VeriHarness (Zhang et al. 2026, preprint) resolves disagreements among one model's attempts with an agentic verifier that "checks competing claims against environmental evidence", and a challenger tests claims the attempts agree on.
- For Task 1.1: re-examining agreed answers is not new; these partial searches found no per-question rule driven by a dependence-aware posterior fitted to unlabeled answers.

Why this method: the math behind it

Result	What it gives this task	Assumptions, and where it fails	Source
Optimal weighted vote, weight log(p/(1 - p)) for accuracy p; bloc rule, one vote per correlated bloc	P1's yes/no cap is this weight at p = 1 - λ, making the bloc rule exact	Independent voters, known accuracies, two options, equal priors; the bloc rule is "a good approximation" when correlations between blocs are small	Nitzan and Paroush 1982; Shapley and Grofman 1984, Theorem II
Multi-class optimal weight log[(K - 1)a/(1 - a)] for accuracy a	P1's equal-likelihood corollary, a = 1 - λ	Errors independent given the truth, also in its label-free OW-L and ISP; equally likely wrong answers	Ai et al. 2026, ICML, Theorem 3.1
Evidence caps: -ln(1 - π) nats from any number of verifier gates sharing a blind spot of mass 1 - π; at most 1/ρ effective judges at average error correlation ρ	Closest 2026 analogues; present P1 as their latent-group version	Han seen at abstract level only; Hossain et al. estimate ρ from trusted labels	Han 2026 (preprint); Hossain et al. 2026 (preprint), Corollary 1
Kruskal uniqueness for hidden-class models	P2	Three groups independent given the truth; a rank condition; positive priors; known groups; up to relabeling	Kruskal 1977; Allman et al. 2009, Annals of Statistics
Reject rule; Bayes-optimal deferral to an expert	Eq. 3, τ = 1 - ε_D - c/L	Calibrated posterior; fixed cost and expert accuracy; expert errors independent of the agents'	Chow 1970; Mozannar and Sontag 2020, ICML

P1 needs only that members depend on the truth through their group's shared hidden answer h, however they depend on one another; a simulation reached the yes/no bound to four decimals (researcher's derivation and computation). The draft's multi-class formula, log[(|Y| - 1)(1 - λ)/λ], also assumes equally likely wrong answers: with three answers, λ = 0.3 and the shared mistake on one wrong answer 29 times in 30, a large, accurate group shifts the log-odds (log ratio of two answers' probabilities) by 4.25 nats, not 1.54. State as P1 the general bound below on how far a group's answers X_g can favor y over y', here log(0.7/0.01) = 4.25, with Ai et al.'s weight as the equal-likelihood corollary:

$$\log \frac{P(X_g \mid y)}{P(X_g \mid y')} \leq \max_h \log \frac{P(h \mid y)}{P(h \mid y')}$$

P2's condition is sufficient by Kruskal's uniqueness theorem as Allman et al. 2009 apply it (Corollary 2), but P2 needs four additions:

1. Identification is only up to relabeling the answers, so Task 2.2's planted cases should anchor which answer is which.
2. Groups must be known or recoverable; exact recovery by best fit is NP-hard, meaning no efficient algorithm is known (Jaffe et al. 2016, Lemma 3), so prefer groups observed through a shared base model or cited source.
3. It identifies each group's answer distribution given the truth, which is all the posterior (each answer's probability given all answers) needs.
4. The condition is stronger than needed: two groups with linearly independent answer distributions (none a weighted sum of the others) plus a third whose distributions differ for every pair of truths suffice.

Eq. 3 is Chow's reject rule and the Bayes-optimal rule for deferring to an expert, here the debate, plus a cost (Chow 1970; Mozannar and Sontag 2020, Proposition 2): debating, with expected loss L ε_D + c, beats deciding now, with L(1 - p), exactly when $p < \tau$ (p: leading answer's posterior; L: loss from a wrong decision; ε_D, c: debate error rate and cost). It needs a calibrated posterior (probabilities that match observed frequencies) and a judge whose errors do not follow the agents' shared mistakes, which is why Task 1.2's judge shares no model with either side.

With $\lambda = 0.3$ per group and a uniform prior, unanimous agents in k groups that err independently reach a posterior of at most 0.700, 0.845, 0.927 and 0.967 for k = 1 to 4, so at $\tau = 0.85$ (ε_D = 0.10, c/L = 0.05) unanimity within one or two groups goes to debate (researcher's computation). Unlike JuryProbe and VeriHarness, this rule needs no labels and applies per question across different agents.

Closest work and how this task differs

Work	What it does	What it assumes	What this task adds
MiniRep, Zhang et al. 2026 (preprint)	Reputation weights in multi-agent discussion; each declared same-model group counts once, with capped weight	Groups given in advance; verifier trained on correctness labels; at most f malicious agents	Group error rates from unlabeled answers; P1; debate decision; audits
JuryProbe, Zhou and Lin 2026 (preprint)	Re-evaluates majority accepts of panels whose judges' false negatives correlate	Labeled probe; trusted references; no formal guarantee	Per-question escalation from unlabeled answers; planted cases only as anchors
CARE, Zhao et al. 2026, ICML	Separates quality from shared confounders, such as style, without labels; up to 26.8% lower error than majority vote	Quality is the dominant shared factor; judges not strategic	Shared mistakes as a hidden answer that can replace the truth; planted anchors; debate decision
Balasubramanian et al. 2026 (preprint)	Label-free Ising (pairwise-interaction) and latent-factor models; proves independence-based aggregation inconsistent under dependence	Binary; item-level factor independent of the truth; identifiability not stated	Multi-class group answers; P1; P2; debate decision
Jaffe et al. 2016, AISTATS	Hidden group variables between truth and classifiers; spectral group detection	Binary; "typically only an approximation"; exact group recovery NP-hard	Multi-class; P1; P2 with observed groups; debate decision

The closest 2026 paper is MiniRep, which already counts each declared same-model group once and resists malicious agents; JuryProbe and VeriHarness already re-examine agreed answers. The safest true claim is the combination: group error rates estimated from unlabeled answers, P1 as the latent-group version of the caps of Han 2026 and Hossain et al. 2026, planted-case anchors, and per-question escalation to debate by a cost threshold. The web-search budget ran out before the August-October 2026 sweep finished, so the novelty sweep should be re-run before submission, starting with the unopened RAIM (arXiv 2609.39229), which cites Kohli and Hossain et al.

What to borrow

Technique	Source	How to adapt it here
Three-group spectral start, then EM	Zhang et al. 2016	Use model families as groups; fit eq. 2 from the spectral start; cite the rate
Anchors to fix which factor is which	CARE, Zhao et al. 2026, Section 3.3	Use Task 2.2's planted cases as anchors
Shared false values as evidence of copying	Dong et al. 2009	Group agents by agreement on answers the posterior deems wrong
Counting sources, not copies	Zanolini 2026; Gao et al. 2026 (preprints)	Group agents by cited documents and tool calls; declare forwarded copies, which cut LLM aggregators' weight on a copy to 0.08 of a new reading or less
All-fail rate (share of items every model misses)	J. Chen 2026 (preprint)	Add a hard-item class, since P2 assumes no item-level common cause
Learning when to defer to an expert	Mozannar and Sontag 2020	Learn ε_D per topic from logged debates and audits

Fixes to the white paper

- **Section 3, first sentence.** It cites Jaffe et al. 2016 for a limitation that paper partly removed, and "never change their minds" fails for WISE (Cherian et al. 2025, preprint, venue unconfirmed), which runs a modified Dawid-Skene across discussion rounds. Suggested: "Statistical methods estimate each rater's accuracy without an answer key (Dawid and Skene, 1979; Whitehill et al., 2009), and some allow dependence: sources that copy one another (Dong et al., 2009), groups of related classifiers (Jaffe et al., 2016), or shared biases among AI judges (Zhao et al., 2026). But each fixes the form of dependence in advance, treats answers as fixed reports from raters that do not act strategically, and none decides when the evidence is too weak and a question needs further scrutiny."
- **Task 1.1 text.** Cite Jaffe et al. 2016 and Dong et al. 2009. Suggested: "The core method extends the rater model of Dawid and Skene (1979) with shared error factors, as Jaffe et al. (2016) did for binary classifiers: agents that keep giving the same unusual answers share a factor (Dong et al., 2009), and their agreement adds little evidence."
- **Simulation (20-page draft).** It reports up to 0.85 for five independent agents at 70% plus k copies of a source wrong 30% of the time. The researcher computed the Bayes ceiling (best achievable accuracy, counting the copy group once) as 0.837 for yes/no questions, and 0.914 or 0.937 with three or four options and wrong answers spread evenly. If questions are yes/no, 0.85 exceeds the ceiling: state the number of options or recheck the figure, and plot the ceiling with confidence intervals.
- **Citations (20-page draft).** CARE is Zhao et al. 2026, separate from Balasubramanian et al. 2026, and assumes quality is the dominant shared factor; DOWN is Eo et al. 2025. Drop the unidentified second Hu et al. 2026 cited with DOWN, or cite Kohli's 9.1% instead, and cite Chow 1970 and Mozannar and Sontag 2020 at eq. 3.

Draft related-work paragraph

> Rater models estimate each source's accuracy without an answer key (Dawid and Skene, 1979). Later work handles copying (Dong et al., 2009), shared biases of AI judges (Zhao et al., 2026) and declared same-model groups (Zhang et al., 2026a). We give each group of dependent agents a shared hidden answer, extending Jaffe et al. (2016) beyond yes/no questions. We show that such a group, whatever its size, adds no more evidence than its shared source. This makes exact the classical advice to count a correlated bloc as one voter (Shapley and Grofman, 1984) and parallels recent caps (Han, 2026; Hossain et al., 2026). Track records are identifiable from answers alone when three known groups err independently and are informative enough (Kruskal, 1977; Allman et al., 2009). Planted cases fix which answer is which. A question goes to debate when its leading answer is too uncertain given the costs, the optimal rule for deferring to an expert (Chow, 1970; Mozannar and Sontag, 2020). Earlier systems escalate on one agent's confidence (Eo et al., 2025) or re-examine agreed answers with labels or within one model (Zhou and Lin, 2026; Zhang et al., 2026b). Ours decides per question without labels and can send even a unanimous answer to debate.

Task 1.2: Impartial choice of debaters and judges

For Task 1.2, which picks each debate's representatives and judge, Proposition P3's impartiality claim (no agent's ratings change its own chance of a role) can be proved in a few lines by the partition argument of Alon et al. 2011, if rater weights and model groups are fixed before each question's ratings. No deterministic impartial rule can guarantee any fixed fraction

of the top candidate's support, so lotteries are forced (Alon et al.), and the best lottery keeps exactly half of it (Fischer and Klimm 2015). **The most important change: because agents may care which answer wins, rate candidates only by same-answer agents outside their model group or by neutral agents, and fix the judge pool before answers are given.**

Prior work and its limits

Impartiality forces lotteries.

- Holzman and Moulin 2013 (Econometrica) define impartiality: an agent's own nominations never change whether it wins. No deterministic impartial rule guarantees any fixed fraction of the top candidate's support, its number of nominations (Alon et al. 2011, TARK, Thm 3.1).
- The best impartial lottery keeps exactly half of that support in the worst case (Fischer and Klimm 2015, SIAM J. Computing), and 2/3 with up to two winners, which is tight (Bjelde et al. 2017). The loss nearly vanishes when the top candidate has many supporters (Bousquet et al. 2014, arXiv, venue unverified) or a good prediction of the winner exists (Cembrano et al. 2025, NeurIPS).
- For Task 1.2: the lotteries follow from the goal, and a co-representative per side or track records used as predictions would lower this price of impartiality.

Partitions and weights in peer review.

- Kurokawa et al. 2015 (IJCAI) study impartial peer review with numeric ratings. Their Credible Subset rule reaches k/(k + m) of honest-review quality, for k winners and m reviews each, but may select nobody; it is not a partition rule.
- Partition rules count only ratings from other groups (Alon et al.). Groups fixed in advance by conflicts keep the rule strategyproof, Alon et al.'s word for impartial (Xu et al. 2019, IJCAI), and block coalitions inside a group (Aziz et al. 2019, Artificial Intelligence, Remark 1). Groups chosen from the reports make it "not strategyproof" (Aziz et al.).
- Weighted PeerNomination (Lev et al. 2023, Artificial Intelligence) weights reviewers by agreement with co-reviewers or panel majorities. Its strategyproofness proof needs special review assignments, it excludes weights fitted by EM (an iterative estimation method) because they break strategyproofness, and "identification becomes impossible" when accurate agents are very few.
- For Task 1.2: model groups act as conflict groups, and Lev et al.'s limits show why weights should come from separate data.

Coalitions call for caps.

- No coalition-proof lottery is much better than a uniform random pick, "even if one is merely interested in coalitions of size at most two" (Alon et al. 2011, Thm 4.4).
- Randomized reviewer assignment caps each reviewer-paper probability "regardless of how extreme a reviewers' manipulation of the assignment is" (Jecmen et al. 2020, NeurIPS). It "does not address the problem of reviewers colluding with each other", and the paper states no coalition bound.
- For Task 1.2: no selection rule can block cross-group collusion without losing nearly all quality, so probability caps and audits (Task 2.2) must handle it.

LLMs favor their own models.

- LLM evaluators recognize and favor their own outputs (Panickssery et al. 2024, NeurIPS), and GPT-4o and Claude 3.5 Sonnet "systematically assign higher ratings to outputs produced by other models of the same family" (Spiliopoulou et al. 2025, preprint).
- Synonym-level perturbations reduce self-preference, but fully neutralizing style brings it back (Mahbub and Feng 2025, preprint). Stronger models prefer themselves "mostly legitimately" (Chen et al. 2025, preprint).
- For Task 1.2: hiding names is not enough, so a group should be a model family, including fine-tuned and distilled descendants, and evaluations should report the accuracy lost by excluding siblings' ratings.

Why this method: the math behind it

Result	What it gives this task	Assumptions, and where it fails	Source
Partition rule counts only other parts' nominations; no deterministic impartial rule guarantees a fixed fraction (Thm 3.1)	P3's proof; lotteries are forced	Agents care only about their own selection	Alon et al. 2011
Best impartial lottery keeps 1/2 of the top support; 2-partition keeps exactly 1/4	Worst-case price of choosing one agent	Approval nominations, abstentions allowed; random order or partition, unlike fixed model groups	Fischer and Klimm 2015
Fixed clusters stay strategyproof and block coalitions within a cluster; clusters chosen from reports do not	Model groups as clusters; Dollar Partition Raffle for the group draw	Equal raters; no guarantee relative to the truth	Aziz et al. 2019
Coalition-proof lotteries do little better than a random pick, even against pairs (Thm 4.4)	Cross-group collusion goes to caps and audits	Agents value only their own selection	Alon et al. 2011
Each reviewer-paper probability is capped, whatever the manipulation	Basis of P3's judge-lottery bound	No coalition bound stated; collusion after assignment not covered	Jecmen et al. 2020
Minimizing the largest chance is optimal against manipulation, up to constants; leximin, the deployed rule, is not	Draw the judge by a min-max lottery	Rich pool; rounding-based algorithms; can give some agents zero chance	Flanigan et al. 2024

The program officer asked how weighting and combining contributors differs from Procaccia's work on impartial peer review and panel selection. Impartial peer review guarantees that no participant's ratings change its own chance of selection, but counts every rater equally (Alon et al. 2011; Kurokawa et al. 2015), and panel selection draws lotteries with chances as equal as possible, showing that keeping the largest chance small limits manipulation (Flanigan et al. 2021, Nature; Flanigan et al. 2024). Task 1.2 combines the two and weights raters by Task 1.1 track records, estimated from answers and never from the current ratings, so no rater can raise its own weight and impartiality holds exactly.

P3's impartiality claim holds as stated when groups and weights are fixed before the question's ratings: the draws ignore ratings and the drawn group's members are ranked only by outsiders, and since agents sharing a model never rate one another, coalitions inside a model group are blocked too (Aziz et al. 2019). Weights may then be fitted by EM, as in Dawid-Skene rater models, which Lev et al. 2023 had to exclude; their agreement-based weights need special review assignments. P3's price claim should change: the known 1/4 and 1/2 ratios assume random partitions, a uniform draw over groups fixed by model guarantees only 1 over the number of groups giving that answer (researcher's derivation, assuming outsiders rank the drawn group's best member first), and draws in proportion to outsiders' ratings (Dollar Partition Raffle) or using track records as predictions (Cembrano et al. 2025) do better when these favor the best group.

P3's judge-lottery bound, below (J is the judge, C the colluders, π_j agent j's chance of judging), is a union bound on the per-pair caps of Jecmen et al. 2020, who do not state it, so present it as the PI's corollary. A min-max lottery minimizes the bound and is the most manipulation-robust lottery (Flanigan et al. 2024), while a draw weighted by track record makes the capture probability equal the colluders' share of weight, which audits must keep small. The bound holds per question, so captures add up across debates: under a uniform draw, one colluder among 20 eligible judges captures at least one of 50 debates with probability 0.92 (researcher's computation).

$$P(J \in C) = \sum_{j \in C} \pi_j \leq |C \cap \operatorname{supp} \pi| \cdot \max_j \pi_j$$

Closest work and how this task differs

Work	What it does	What it assumes	What this task adds
MiniRep, Zhang et al. 2026 (preprint, Sept)	Reputation-weighted vote with no judge or theorem; declared same-model groups count once, with capped weight; top 7 of 10 agents by weight	Clone map given; verifier trained on correctness labels; at most 3 of 10 malicious	Roles drawn by lottery, with a proof; a judge; groups that can be estimated; audits
PeerRank, Margalit et al. 2026 (preprint, Feb)	12 models write, answer and judge tasks; score is an unweighted mean excluding self-ratings	Equal judges; no strategic behavior, so no impartiality notion or collusion analysis	Sibling exclusion; track-record weights; lotteries; a collusion model
Auto-Arena, Zhao et al. 2025 (ACL)	Judges are the top 5 models by current Elo, a rating from pairwise wins, excluding participants and their families	Current ranking is trustworthy; no collusion	Capped lotteries; a judge who rates no one; weights judges cannot move
PiCO, Ning et al. 2025 (ICLR)	Learns reviewer weights consistent with the scores each model receives	Good answerers are good reviewers	Weights independent of peer scores; impartiality
PRD, Li et al. 2023; PeerEval	Weights each reviewer by how well its own answers are judged; self-judgments included	Better contestants are better reviewers	No self or sibling ratings; track-record weights; ratings pick only who argues
Lev et al. 2023 (AIJ)	Weighted PeerNomination; weights from agreement with co-reviewers	Accurate majority; special assignments; single deviations	Weights from answers and planted cases; model groups; a judge

The closest 2026 paper is MiniRep, which cites no impartial-selection work, and the safest true claim is that, to our knowledge, no LLM work picks debaters or judges impartially. The search was partial, because the web-search budget ran out and the August-October 2026 scan covered arXiv cs.GT and cs.MA by title only, so the novelty sweep should be re-run before submission, including PeerRank's later version, which Semantic Scholar lists as citing Kurokawa et al. (v1 does not).

What to borrow

Technique	Source	How to adapt it here
Dollar Partition Raffle	Aziz et al. 2019	Draw each side's group in proportion to outsiders' normalized weighted ratings, not uniformly
Predictions with a tunable level ρ	Cembrano et al. 2025	Use track records as the prediction for the group draw; tune ρ to trade the guarantee when records are right against the guarantee when they are wrong
Probability caps by linear program and sampling	Jecmen et al. 2020	Cap each agent's chance of any role at Q; bar the judge from either side's family; state $P(\text{capture}) \le f \cdot Q$
Min-max lottery	Flanigan et al. 2024	Choose the judge lottery that minimizes the largest chance under the exclusions
Transparent lottery	Flanigan, Kehne and Procaccia 2021	Publish m candidate role assignments and draw one with public randomness, so audits can confirm the draw
Cycle-free rating assignment	Boehmer et al. 2022	Keep two groups from rating each other's candidates; in peer review, forbidding such pairs cost at most 0.8% of assignment weight on ICLR-18 data
On-off attacks; group weight caps	MiniRep	Cap each family's total rating weight; red-team with agents that build a record, then attack

Fixes to the white paper

- **Section 3, impartial peer review and Lev et al.** Credit the term "impartial" to Holzman and Moulin 2013. The Lev et al. sentence is essentially right, but should name what the weights measure and stay consistent with Task 1.1, whose track records also rely on agreement. Suggested: "Lev et al. (2023) weight reviewers by agreement with the other reviewers of the same candidates, which, as they note, cannot identify accurate reviewers when these are few. Our track records also use agreement; planted cases and counting shared mistakes once are what let them identify accurate minorities."
- **Section 3, "take opinions as given".** Too broad: peer grading already spot-checks raters against partial ground truth (Gao, Wright and Leyton-Brown 2016, preprint). Suggested: "These mechanisms treat ratings as unverifiable and guard only against self-promotion; in our setting, disputed steps can be re-run and planted cases anchor track records."
- **Task 1.2, partition citation.** Kurokawa et al.'s Credible Subset is not a partition mechanism. Cite Alon et al. 2011 with Aziz et al. 2019 or Xu et al. 2019 for the partition idea, and Kurokawa et al. for the peer-review setting.
- **Task 1.2, "as in our PeerEval leaderboard and PRD method".** PRD weights each reviewer by how well its own answers are judged and includes self-judgments, so it is not impartial. Suggested: "extending PRD, whose weights come from peer-judged scores; here weights come from track records estimated without peer ratings."
- **Task 1.2, "rated only by outsiders".** The opposing side is also outside: it can rate a side's weakest arguer highest, and a candidate that rates the other side can choose its own opponent. Suggested: "rated only by agents outside its model group that gave the same answer, or by neutral agents." A same-answer colluder can still push a weak representative, which Task 2.2's audits must catch.
- **Task 1.2, "a random agent that did not answer".** Colluders can abstain to become eligible as judge, which P3's judge-lottery bound assumes cannot happen. Suggested: "The judge is drawn from a pool fixed before answers are given, with each agent's chance of judging capped."
- **Section 4, "the first leaderboard on which AI models rank one another".** Not supported: Language-Model-as-an-Examiner (Bai et al. 2023) ranked four models by mutual examination in June 2023, a month before PRD. Suggested: "PRD (2023) was among the first methods in which AI models rank one another; our PeerEval leaderboard applies it to current models."

Draft related-work paragraph

> Impartial selection asks that no participant's ratings change its own chance of being chosen (Holzman and Moulin, 2013), as in impartial peer review (Kurokawa et al., 2015). No deterministic rule does this well, so impartial rules use lotteries (Alon et al., 2011). The best lottery keeps half of the top candidate's support in the worst case (Fischer and Klimm, 2015). Groups fixed in advance and rated only from outside keep impartiality (Xu et al., 2019) and stop collusion within a group (Aziz et al., 2019). Lev et al. (2023) add reviewer weights based on agreement, whose impartiality needs special review assignments. LLM peer evaluations exclude self-ratings (Margalit et al., 2026) or judges from the contestants' model families (Zhao et al., 2025). MiniRep counts declared same-model groups once (Zhang et al., 2026). None of these picks debaters or judges impartially. We group agents by underlying model and weight raters by track records estimated from answers, so impartiality holds exactly on every question. Following capped assignment lotteries (Jecmen et al., 2020) and panel lotteries (Flanigan et al., 2024), we draw the judge so that f colluders hold the seat with probability at most f times the largest individual chance.

Task 2.1: Debate before an impartial judge

In the work we found, no debate judge re-runs the exact tool calls that debaters cite, as Task 2.1's impartial judge does at the first disputed step. Refereed delegation of computation, a proved and deployed protocol, justifies both the stop and the re-run (Canetti et al. 2011, CCS; Canetti et al. 2013, Information and Computation). When two servers disagree, the client bisects (halves the disputed range each round) to the first differing step, recomputes only that step, and gets the right result in logarithmically many rounds if one server is honest. **The most important change is to recast the stopping rule as refereed delegation: both representatives commit their steps and tool logs on one shared outline of sub-questions, and the loop stops at the first sub-question on which they differ.**

Prior work and its limits

Discussion erases disagreement. In Shu 2026 (preprint), seven LLM judges rated 600 code-correctness candidates, revised after seeing one another's answers, then voted; unanimity rose from 39.5% to 95.2% after one round, while accuracy moved by less than one point. The PI's PRD peer discussion also seeks agreement, with no judge (Li et al. 2024). For Task 2.1: let the judge's choice decide, and have each side commit before seeing the other's argument.

Judges read rather than verify. In standard debate, judge accuracy rose from 48% to 76% for non-expert models and from 60% to 88% for humans, with quotes matched against the source (Khan et al. 2024, ICML); Kenton et al. 2024 (NeurIPS) leave tool use and code execution to future work. Judges' verification fell once a critic appeared (gpt-oss: from 67–82% to 16–31%; Elasky et al. 2026, preprint), and an LLM auditor adopted agents' stated conclusions 94.4% of the time, against 3.4% once that field was removed (Arslan 2026, preprint). For Task 2.1: the harness, not the judge's diligence, should re-run the cited tool calls.

Debate theory assumes an honest side. Irving et al. 2018 (preprint) prove that with optimal play, debate with a polynomial-time judge can decide any problem in PSPACE (solvable with polynomial memory), but call such arguments "analogies only". Barnes and Christiano 2020 (Alignment Forum) found obfuscated arguments, long arguments whose few flawed steps neither side can locate, for every physics question studied, and reported no fix. For Task 2.1: since LLMs largely cannot find reasoning errors but can correct them given the location (Tyen et al. 2024, Findings of ACL), let the opponent locate the disputed step, and state the guarantee conditionally (correct unless a named assumption fails), as a debate safety case does (Buhl et al. 2025, preprint).

Re-running needs reproducible evidence.

- What breaks it: LLM outputs vary even at temperature 0, because results depend on the server's batch size, which varies with load; batch-invariant routines made 1,000 runs of Qwen3-235B identical, at about 20% slower matrix multiplication (He 2025, blog post). Floating-point order also differs across hardware, and live APIs and web results change.
- Remedies others use: a fixed floating-point order, which lets Verde apply refereed delegation to ML programs (Arun et al. 2025, preprint); cached and simulated APIs (StableToolBench, Guo et al. 2024, preprint); and proofs that data came from a given secure (TLS) website (DECO, Zhang et al. 2020, ACM CCS).
- For Task 2.1, declare four evidence classes (our suggestion): deterministic tools, re-run exactly; archived sources, re-run against the logged snapshot; live sources, logged with call, time, response and hash; LLM calls, pinned to a model version at temperature 0, else treated as live. A live re-run that differs shows that the source changed, not fraud, so the judge rules as of the logged time or reports the step unsettled, as with the vessel example's port report.

Why this method: the math behind it

Result	What it gives this task	Assumptions, and where it fails	Source
Refereed delegation: one honest server suffices; logarithmically many rounds; one step recomputed	The stopping rule; an exact re-run at that step	Deterministic steps, a common input, one honest server, a secure hash. Fails: no shared program without an outline; inference steps cannot be recomputed; shared mistakes	Canetti et al. 2011; Canetti et al. 2013; mechanics read via Ye et al. 2022, Nehab et al. 2024
Fraud proofs (blockchain dispute games): bisection to one virtual-machine instruction, then a one-step proof; TrueBit adds forced errors	A deployed precedent; forced errors keep verifiers alert, like Task 2.2's planted cases	Determinism; at least one honest verifier; timely moves. Fails: nondeterministic tools; timeouts that block the honest side	Kalodner et al. 2018 (USENIX Security); Teutsch and Reitwießner 2019 (preprint)
Refereed games: one round captures exactly PSPACE, polynomially many rounds exactly EXP (exponential time)	The ceiling for competing provers before a fast referee	Unbounded provers, an honest one playing optimally. Fails: bounded debaters	Feige and Kilian 1997 (STOC), read via Kol and Raz 2014
Doubly-efficient debate: O(log T) rounds and O(1) judge queries for a T-step computation (Thm 5.1); one side posts its transcript, the other names a step (Thm 5.3)	The AI-debate counterpart	A shared program M; provers that predict the judge accurately. Fails: obfuscated arguments; a fallible judge	Brown-Cohen et al. 2024 (ICML)

Result	What it gives this task	Assumptions, and where it fails	Source
Blackwell informativeness (at least as useful for every decision): voting and debate, in the paper's terms, are no more informative than the agents' pooled private information	Debate must add evidence, such as re-run tool output	Read from the abstract only; added outside evidence is not covered (our inference)	Zhang et al. 2026a (preprint)
Grounded semantics (the most cautious in, out or undecided labeling of arguments): argument games are sound and complete for it	A meaning for the losing rule and "unsettled"	Attacks set by the judge's rulings. Fails: arguments that add up independent evidence	Dung 1995 (AIJ); Prakken 2006 (KER)

Lean on refereed delegation for P4, which holds as stated if its assumptions include that every argument for a false answer contains a false step. We suggest a sharper form (our synthesis, not a published result), with M the event that the correct side disputes none of the wrong side's false steps, B the event that it argues a false step itself, and ε_i the judge's error on disputed step i. A wrong decision needs the wrong side to escape (M or a misruling) and the correct side to be struck (B or a misruling); one failure alone yields "unsettled".

$$P(\text{wrong decision}) \le \min\left\{P(M) + \sum_{i \in \text{wrong side}} \varepsilon_i,\ P(B) + \sum_{j \in \text{correct side}} \varepsilon_j\right\} \le P(M) + P(B) + \sum_{\text{all disputed } k} \varepsilon_k$$

The right-hand side is P4's union bound; with a shared outline only one step is disputed, so it shrinks to the chance that the correct side errs there plus the judge's error on it. Refereed delegation makes all three terms zero, up to hash collisions: determinism and a common input give P(M) = 0, an honest server P(B) = 0, and exact recomputation ε = 0. Each assumption that fails for LLM agents adds its term, and one part of the project targets each: Tasks 1.1 and 1.2 target P(B) and shared mistakes, the outline and stopping rule P(M), re-running ε on tool-backed steps only, and Task 2.2 thrown debates and true but selectively chosen facts, which re-running cannot catch (Hu et al. 2026, ACL).

For P4's losing rule, lean on grounded semantics, with steps as arguments and disputes as attacks: a move is out if some reply to it is in, and in if all replies attacking it are out (Prakken 2006, after Dung 1995). A step the judge cannot rule on leaves both sides undecided, a principled meaning for "unsettled"; the draft should also call it unsettled when both sides have a struck step, unless a burden of persuasion declared in advance sends ties against one side (Prakken and Sartor 2007, ICAIL). Because one struck step sinks an argument, the rule breaks for arguments that add up independent evidence, where prover-estimator debate's probabilities fit better (Brown-Cohen et al. 2025, preprint).

Closest work and how this task differs

Work	What it does	What it assumes	What this task adds
VeriHarness (Zhang et al. 2026c, preprint)	A verifier agent with evidence tools settles disagreements among one model's rollouts (repeated attempts) and probes agreed claims; +6.2 and +6.4 points on two frontier models	One generator; no collusion model; abstract silent on re-running tool calls	Possibly dishonest debaters; exact re-runs of the calls they cite; a stop at the first disputed step
Jiang et al. 2026 (ICML)	Two consultants name disputed premises or steps until they agree or hit a turn limit; a passive judge reads the transcript; 62.1% vs 49.2% for standard debate with weak judges	Consultants that "produce reliable reasoning"; at least one correct; no tools	Possibly dishonest representatives; a judge who re-runs calls and decides without consensus

Work	What it does	What it assumes	What this task adds
PROClaim (Chowdhury et al. 2026, preprint)	Courtroom roles; an LLM arbiter admits evidence from a fixed index; 3-judge majority; 81.7% on Check-COVID vs 85.8% for one GPT-5-mini call with retrieval	Judges do not re-retrieve; INCONCLUSIVE counts as SUPPORT	Exact re-running; an impartial judge; an "unsettled" output
Tool-MAD (Jeong et al. 2026); DynaDebate (Li et al. 2026) (preprints)	Tool-MAD stops on consensus and its judge calls no tools; DynaDebate runs tools only "when a deadlock persists"	Consensus, extraction or a vote decides	The opponent names the disputed step; the judge re-runs the exact calls against logged snapshots
Prover-estimator debate (Brown-Cohen et al. 2025)	Theory: an estimator gives probabilities for subclaims; the judge rules on one leaf; no LLM version	Stability; equilibrium; a correct leaf judge	An implemented harness with tools

The closest 2026 papers are VeriHarness and Jiang et al., and standard debate confirms only quotations (Khan et al. 2024); the safest true claim is that, in the work we found, no judge re-runs the exact tool calls that debaters cite, in a two-sided debate stopped at the first disputed step. The web-search budget ran out before a full August–October 2026 sweep, so the search was not exhaustive and the novelty sweep should be re-run before submission.

What to borrow

Technique	Source	How to adapt it here
Commit, then find the first differing step	Canetti et al. 2013; Teutsch and Reitwießner 2019	Both sides commit steps and a hash-chained (tamper-evident) tool log; bisect only inside long tool traces
Shared outline	Jiang et al. 2026 (collaborative phase)	A short collaborative phase or a testbed template fixes the sub-questions
Consensus challenger	Zhang et al. 2026c	Probe steps both sides agree on, which the stopping rule never visits
Product-of-posteriors baseline	Zhang et al. 2026a	It beat voting and debate on six QA benchmarks; ours should gain where re-run evidence adds information
Step parsing of search calls	HiPRAG (Wu et al. 2026, ICLR)	Each parsed search call becomes a logged, re-runnable evidence unit

Fixes to the white paper

- **Shu 2026.** Shu does not say systems "discuss until they agree"; its panels revise for 1–3 rounds, then vote. Suggested: "In multi-agent discussion, agents read one another's answers and revise them; a single round can make a panel unanimous without making it more accurate (unanimity rose from 39.5% to 95.2% while accuracy moved by less than one point), erasing the disagreement that exposes errors (Shu 2026)."
- **Stopping-rule citation.** Irving et al. do not define "the first step on which they differ". Suggested: "the two answer the same sub-questions in order, and the loop stops at the first one on which they differ, as in refereed delegation of computation, where a client settles a dispute between two servers by recomputing one step (Canetti, Riva and Rothblum 2013; Brown-Cohen, Irving and Piliouras 2024)."
- **Irving claim ("Why it will succeed").** Open with refereed delegation and fraud proofs, as in the draft paragraph below, then add "with optimal play" and the obfuscated-arguments caveat: "In debate with optimal play, a judge who rules on one step can settle questions far beyond its own ability (Irving et al. 2018; Brown-Cohen et al. 2024). These results assume an honest side that can find the flawed step, which can fail for long arguments (Barnes and Christiano 2020), so our judge may report a ruling as unsettled."
- **Kenton et al. 2026.** Its headline is that debate training reduced reward hacking; invented quotes appeared only with a deliberately weakened judge (Kenton et al. 2026, preprint). Suggested: "even debate training, which reduced reward hacking overall, taught a critic to invent quotations when the judge was weakened (Kenton et al. 2026)."

- **"Fixed rounds of text".** This overstates. Suggested: "Most LLM discussions run a set number of rounds, or stop once the agents agree, among a fixed set of models; some retrieve documents or run code, but their judges read the transcript and do not reproduce the tool calls (Jeong et al. 2026; Chowdhury et al. 2026)."
- **Agent-as-a-Judge.** Its best system has no execution module (Zhuge et al. 2025, ICML), so re-running is the new part. Suggested: "the judge, an agent like those that already evaluate other agents' work (Zhuge et al. 2025), re-runs the call in the sandbox."
- **"No existing debate protocol holds up when agents share mistakes, lie, or collude."** This overreaches. Byzantine-robust consensus among LLM agents, which tolerates members that fail in arbitrary ways, already bounds the number of faulty agents it can absorb (Lee et al. 2026, ICML workshop), and MiniRep's reputation weights resist malicious agents in discussion (Zhang et al. 2026b, preprint). Suggested: "Existing defenses bound the number or influence of faulty agents (Lee et al. 2026; Zhang et al. 2026b); none that we found also counts honest agents' shared mistakes once or audits the judge."
- **Bibliography.** Zhang et al. 2026a, b and c are the Blackwell, MiniRep and VeriHarness papers. Check Feige and Kilian 1997 and the CCS 2011 mechanics in the originals before citing details.

Draft related-work paragraph

> In multi-agent discussion, one round of reading and revising can make a panel unanimous without making it more accurate (Shu, 2026). Neither voting nor such exchange yields a decision more informative than the agents' pooled private information (Zhang et al., 2026a), so a debate must add evidence to do better. Standard debate adds quotations matched against a source text (Khan et al., 2024; Kenton et al., 2024). Other systems retrieve documents or settle disagreements with evidence tools (Jeong et al., 2026; Zhang et al., 2026c), but in the work we found, no judge reproduces the tool calls that debaters cite. Our stopping rule follows refereed delegation of computation. When two servers disagree, a client recomputes only the first step on which they differ; if one server is honest, the result is correct (Canetti et al., 2013). Blockchain fraud proofs use this method (Kalodner et al., 2018), and debate theory gives its counterpart for AI (Brown-Cohen et al., 2024). Our judge likewise re-runs the tool calls at the disputed step. These results assume an honest side that can find the flawed step, which can fail for long arguments (Barnes and Christiano, 2020), so our judge may report a ruling as unsettled.

Task 2.2: Automatic and human audits against collusion

At 10% honest error and a 2% audit rate, a colluding debater or judge can steer about 930 of 10,000 decisions before Task 2.2's audits, on planted cases and a few random real decisions, are 80% likely to flag it (the researcher's calculation). In AI control, the closest prior work, only an attacking model produces the bad outputs being audited, so one caught output proves an attack; Griffin et al. 2024 (presented at IASEAI'26) name this assumption. Inspection games justify random, committed audits: an inspector who commits in advance to random inspections keeps the inspectee legal while inspections remain, and an inspectee who sees the schedule times its violation (Dresher 1962, RAND; Maschler 1966, Naval Research Logistics Quarterly; von Stengel 2016, Mathematics of Operations Research). **The most important change is to split P5: keep (1 - ρ)^n as a bound on audit coverage under stated conditions, and add a detection limit that grows with the square root of the number of decisions.**

Prior work and its limits

Predictable audits are dodged. Auditing the first 2% of programming outputs gave only 15% safety (the chance that no backdoor got through), because the attacker inserted backdoors rarely (Greenblatt et al. 2024, ICML). When a trusted monitor sent its most suspicious outputs to audit, choosing when to attack cut safety from 99% to 59% at a 0.5% budget (Schaeffer et al. 2026, preprint). LLM agents asked to pick about 1% of requests for human review at random chose predictable ones: in a follow-up, GPT-6 Sol selected 17 of 1,008, including 17 of the 18 identifiers below 1% of the value range (Ezaki et al. 2026, preprint). For Task 2.2: trusted code, never an agent, draws each audit after the decision is logged.

Planted cases can be recognized.

- Colluding crowd workers spotted reused gold questions (tasks with known answers) by how often each recurred. In a modeled 2,000-task job with 5% gold, detection exceeded 99.9% once 50 workers had joined (Checco et al. 2018, HCOMP).
- Gemini-2.5-Pro told evaluations from real use with AUC 0.83 (0.5 is chance), against 0.92 for humans, and more easily in agentic settings (Needham et al. 2025, preprint).
- Models also act differently when they suspect an evaluation. Claude 3 Opus complied with harmful requests 14% of the time for users whose conversations it was told were used for training, and almost never for others (Greenblatt et

al. 2024b, preprint); evaluation awareness causally lowered covert behavior in Schoen et al. 2025 (preprint).
- For Task 2.2: build planted cases fresh for each model family by minimal edits of real cases. Random audits of real decisions cover colluders that still recognize them.

Honest reviewers miss planted errors. BMJ reviewers found on average 2.58 of 9 major errors planted in their first paper (Schroter et al. 2008, JRSM). On 91 real errors that led to errata or retractions, the best model, o3, had 21.1% recall and 6.1% precision (Son et al. 2025, preprint). For Task 2.2: honest misses weaken any flag, so set predicted error rates per error type and difficulty, or honest agents that miss hard errors will be flagged.

Multi-agent defenses bound the bad share. Self-Anchored Consensus states its guarantees as (F+1)-robustness conditions on the communication graph, which bound the number of faulty agents (Lee et al. 2026, ICML workshop). A panel of LLM judges aggregated by the geometric median, a multidimensional median, tolerates at most half corrupted judges (Acharya et al. 2026, preprint), and MiniRep, which weights agents in discussion by reputation, wins all 28 attack conditions with at most 3 malicious agents among 10 (Zhang et al. 2026, preprint). For Task 2.2: none audits against ground truth, and their guarantees need a bounded share of bad agents, a bound that trusted audits can lift.

Why this method: the math behind it

Result	What it gives this task	Assumptions, and where it fails	Source
Inspector leadership: committing in advance to random inspections keeps the inspectee legal while inspections remain; an inspectee who sees the schedule times its violation	Announce the audit rate and flag rule; draw audits secretly, after each decision (P5a)	Rational inspectee, known payoffs, perfect detection; fails for misgeneralized agents and imperfect audits	Dresher 1962; Maschler 1966; von Stengel 2016, Thm 5
Security and audit games: commitment loses nothing even if the attacker cannot observe it; audit rates and punishment set jointly	Spread audits; pair the audit rate with the weight cut after a flag	Known payoffs; fails with several attacker resources, that is, a colluding group	Korzhyk et al. 2011, JAIR; Blocki et al. 2013, IJCAI; Blocki et al. 2015, AAAI
Tax audits and spot-checks: declared income rises with audit probability; truthful effort is dominant once the audit rate reaches effort cost over reward gap	A minimum audit rate $\rho \geq G/(dL)$, for gain G, detection d, penalty L	Known gains; penalty capped at exclusion; targeted audits pay only on features agents cannot move	Allingham and Sandmo 1972; Reinganum and Wilde 1985; Border and Sobel 1987; Gao et al. 2019
Byzantine agreement: with oral messages, solvable if and only if more than two-thirds are loyal; with unforgeable signatures, for any number of traitors	Honest-majority defenses cannot cover a dishonest majority; trusted audits can	Audits are sampled, so guarantees become statistical	Lamport et al. 1982, ACM TOPLAS
Trusted ratings: a number independent of the item count selects nearly the best items for any reliable share $\alpha > 0$; without them, copiers are indistinguishable when $\alpha < 1/2$	Why audits are needed, and why few suffice	Reliable raters err independently; selection only; per-rater work grows as $1/\alpha^3$; fails for shared mistakes and recognizable planted cases	Steinhardt et al. 2016, NeurIPS
Ville's inequality: a nonnegative wealth that cannot grow on average, starting at 1, ever reaches $1/\alpha$ with probability at most α	P5b's anytime-valid flag at K/α for K agents	Predicted rates fixed before each audit; fails if they use the current audit or are too low	Ramdas et al. 2023, Statistical Science; Shafer 2021, JRSS-A

Byzantine agreement without signatures needs more than two-thirds loyal agents, and robust aggregation tolerates at most half. Trusted audits act like unforgeable signatures, which is why the guarantee of Steinhardt et al. survives a dishonest majority; this answers a panel asking why Byzantine-tolerant consensus is not enough. The guarantee makes the white paper's line 103

accurate but incomplete: it covers selecting nearly the best items, neither deciding each item nor naming dishonest agents; it assumes honest raters err independently, the assumption Task 1.1 says fails; and per-rater work grows as $1/\alpha^3$.

Its trusted ratings fall on random items the adversary cannot single out. Planted cases therefore count only if colluders cannot recognize them, while random audits of real decisions qualify by construction, a theorem-based reason for the human audit stream. Task 2.2 is thus an inspection game with many inspectees who may collude and also err honestly; we found no such game in the literature (search not exhaustive), so it is a candidate theory contribution.

P5 holds once its conditions are stated. Its flag tracks a betting wealth per agent, where X_s marks an error on the agent's s-th audited decision and p_s is its predicted error rate, fixed before that audit:

$$M_t = \prod_{s \le t} (1 + \lambda_s (X_s - p_s)), \qquad 0 \le \lambda_s \le 1/p_s, \qquad \text{flag when } M_t \ge K/\alpha$$

- Coverage (P5a). "(1 - ρ)^n" is an upper bound, exact only for an agent that commits in advance to n steers. It holds for any agent strategy only if trusted code draws each audit after the decision is logged, independently of what agents know; since LLM agents pick predictable "random" decisions (Ezaki et al. 2026), agents must never draw audits. Use ρd if audits catch a wrong decision with probability d; one decisive steer escapes with probability 1 - ρ, so high-stakes decisions need Task 2.1's re-running.
- Flag (P5b). By Ville's inequality, an agent that errs no more than predicted is ever flagged with probability at most α/K, so predicted rates should be conservative, such as Task 1.1's upper confidence bounds. Admissible procedures of this kind (none beats them everywhere) must be built from such wealth processes, so the construction is canonical (Ramdas, Ruf, Larsson and Koolen, venue unconfirmed). In the researcher's simulation (10 agents, α = 0.05, predicted error 0.1), an honest agent was ever flagged within 2,000 audits with probability 0.0037, below the 0.005 bound, and 20% steering was flagged after a median of 41 audits.
- Planted versus real. This flag is weaker. In the researcher's simulation it needs about 100-200 matched pairs to catch 20% steering at α = 0.005 (flag rates 0.569 at 100 pairs, 0.942 at 200), and it keeps its α guarantee only if planted cases match real ones in difficulty, for example as minimal edits of real cases, or by comparing each agent's gap with the median gap across agents.

P5 lacks a detection limit (P5c). The researcher's own calculation gives one; it is a candidate proposition and has not been published. With honest error p0 and audit rate ρ, a colluder among T decisions is flagged with good probability only after about k* steers, where z_α and z_β are standard normal quantiles for the false-flag rate and the miss rate:

$$k^* \approx (z_\alpha + z_\beta) \sqrt{\frac{T\, p_0}{\rho\,(1 - p_0)}}$$

With an exact binomial rule at α = 0.005 and 80% power, T = 10,000 needs about 930 steers (9.3%) at ρ = 2% and p0 = 10%, against 80 when p0 = 0. At those rates T = 2,500 needs about 520 (20.8%) and T = 40,000 about 1,750 (4.4%), an analog of the square-root law of covert communication (Bash et al. 2013, IEEE JSAC).

Steering inside the predicted error rate escapes audits entirely. An agent with a weak track record can place all its predicted errors where it wants, a total influence of at most about 0.278 log-odds (log ratio of two answers' probabilities) per decision, reached at p0 ≈ 0.22. Only its low weight limits it, with Task 1.1 counting shared mistakes once (P1); this supports the draft's weak-record clause, and the proposition should state this cost plainly.

Closest work and how this task differs

Work	What it does	What it assumes	What this task adds
MiniRep (Zhang et al. 2026); Byzantine-robust consensus (Lee et al. 2026; Acharya et al. 2026)	Reputation-weighted discussion; filtering or robust aggregation	A bounded bad share (3 of 10; at most half); no ground truth	Ground-truth audits in place of an honest majority; an audited judge

Work	What it does	What it assumes	What this task adds
Who Audits Whom (Ghanem 2026, preprint)	Grades auditor independence by controller, shared model family and evidence; in simulation an internal audit surfaces 5.9% of the faults it could see	One agent audited at a chosen grade; no random draws or planted faults described	Random and planted audits with statistical flags; its independence argument for human audits
Gardner-Challis et al. 2026 (preprint; NeurIPS 2026 workshop)	Safety case for an untrusted monitor across four collusion strategies; 26.4% safety against a monitor that recognizes its own model's outputs, 90.2% with a trusted monitor and summaries	Planted attacks indistinguishable from real outputs; one generator-monitor pair; a slack of missed honeypots of uncertain size	Human audits of real decisions; a planted-versus-real gap; a flag with a bounded false-flag rate
Schaeffer et al. 2026	Attack selection against suspicion-based audits: safety 99% to 59% at a 0.5% budget	One decisive attack; independent single-step tasks	Uniform draws after the fact; repeated steering; statistical flags
Wen et al. 2024 (preprint)	Bayesian credence over a model's misbehavior rate picks protocols; 80% fewer backdoors than non-adaptive baselines	Known monitor-score distributions; one untrusted model	Audits against ground truth; anytime-valid flags; two audit streams
AI control (Greenblatt et al. 2024; Griffin et al. 2024)	Trusted and untrusted monitoring; upfront auditing 15% safety; a formal control game	One untrusted model; only attacks produce backdoors, so catching one suffices	Honest errors; many agents in debate roles; a human audit stream

The closest 2026 paper is Gardner-Challis et al. 2026, which assumes planted attacks cannot be told from real outputs and, like the other control studies here, treats one caught bad output as proof of an attack, so the honest-error setting is new. Within the searches run, no LLM work combines planted-case audits with random audits of real decisions or compares the two streams; the safest true claim is: "To our knowledge, no prior work audits the outcomes of multi-agent debates with planted cases and random human audits compared against each other, with flags that feed agent weights and role eligibility." The search was not exhaustive, because the web-search budget ran out before the August-October 2026 sweep, so the novelty sweep should be re-run before submission.

What to borrow

Technique	Source	How to adapt it here
Survival formula for planted attacks	Gardner-Challis et al. 2026	A judge's chance of passing n planted debates
Minimal edits of real inputs; redaction; paraphrase	Greenblatt et al. 2024; Shlegeris 2024 (blog post)	Planted cases edited from real reports or papers; paraphrase both streams before the judge sees them
Fresh, non-reused gold with balanced labels	Checco et al. 2018; Le et al. 2010 (workshop)	One-time planted cases per model family; label rates matched to real decisions
Trusted randomness, a fresh draw per decision	Ezaki et al. 2026	A cryptographic generator with a secret seed draws after logging; publish only ρ
Averaging betting scores (e-values); e-BH, a false-discovery procedure for them	Vovk and Wang 2021; Wang and Ramdas 2022	Merge the two flags without splitting α; bound the honest share among flagged agents, under any dependence
Auditors see raw evidence, not conclusions	Arslan 2026 (preprint)	Human auditors see the evidence before the debate's decision

Fixes to the white paper

- **Line 103 (Steinhardt et al.).** Accurate but incomplete. Suggested: "Without trusted ratings, dishonest raters that copy honest ones cannot be told apart from them once honest raters are a minority; with them, a number of trusted

ratings that does not grow with the number of items suffices to select nearly the best items, even when most raters are adversarial, provided honest raters err independently (Steinhardt et al., 2016)."

- **P5 (draft).** Suggested: "(a) If trusted code draws each audit after the decision is logged, independently of what agents know, then whatever an agent's strategy, the chance that it steers n decisions and none is audited is at most (1 - ρ)^n. (c) Because honest agents also err, a colluder is flagged only once its steered decisions exceed a constant times the square root of Tp0/ρ; steering that escapes grows only as the square root of the number of decisions." Keep the anytime-valid flag as (b).
- **Line 91, tax audits.** The analogy is uncited. Suggested: "As in inspection games and random tax audits, the audit rate is announced but the draws are secret (Maschler, 1966; Avenhaus et al., 2002; Allingham and Sandmo, 1972)," citing Avenhaus et al. 2002, a Handbook of Game Theory chapter.
- **Line 91, behavior change.** Needham et al. 2025 show detection only. After "tell evaluations from real use", add "and behave differently when they suspect one (Schoen et al., 2025; Greenblatt et al., 2024b)".
- **Line 101, "expose".** Overreaches, since steering below the detection limit escapes. Suggested: "…and a few human audits of random real decisions that limit how often colluding debaters or judges can steer real decisions, even those behaving well only in planted debates."
- **Line 73.** "No existing debate protocol holds up when agents share mistakes, lie, or collude" overreaches: MiniRep, Self-Anchored Consensus and CP-WBFT (Zheng et al. 2025, preprint) report robustness to malicious agents in LLM discussion. Suggested: "Existing defenses bound the number or influence of faulty agents and rely on agreement among them; none also counts honest agents' shared mistakes once, or audits the judge against outside evidence."
- **Unverified citations.** Drop Al Dahik et al. 2025, cited in the earlier review but never identified, unless it is found. Leave out the Baxt et al. 1998 figures, which come from secondary slides (unverified).

Draft related-work paragraph

> AI control audits an untrusted model, one that may attack on purpose, with a small budget of trusted labor. Auditing its first 2% of outputs gave only 15% safety (Greenblatt et al., 2024). Planted cases help only if models cannot tell them from real ones (Gardner-Challis et al., 2026). Capable models often can, and they act differently when they suspect an evaluation (Needham et al., 2025; Schoen et al., 2025). Control studies assume that only an attacking model produces the bad outputs being audited (Griffin et al., 2024). Our honest debaters and judges also err, so we flag an agent only when its audited errors exceed the rate its track record predicts. A betting procedure keeps the chance of ever flagging an agent that errs no more than predicted below a set level (Ramdas et al., 2023). As in inspection games, the audit rate is announced, and trusted code draws each audit after the decision is logged (Maschler, 1966). Byzantine agreement needs more than two-thirds loyal agents unless messages carry unforgeable signatures (Lamport et al., 1982). Human audits of random real decisions play a similar role. With such trusted ratings, a number that does not grow with the number of items suffices to select nearly the best items, even when most raters are adversarial, if honest raters err independently (Steinhardt et al., 2016).

Task 3: Rewards for contributions to trustworthy decisions

In Task 3, which trains the agents by reinforcement learning (RL) on the group's decisions, every label-free reward the search found has an optimum or fixed point that does not depend on correctness. A model rewarded for matching its own majority vote gave one template answer to nearly every prompt after 1,200 steps (Shafayat et al. 2025, preprint). The classical result behind the method is peer prediction: rewarding agreement with a reference answer, minus the same answer's agreement with the reference on another question, pays zero on average to answers that ignore the question and cancels the reference's bias (Dasgupta and Ghosh 2013; Liu and Guo 2020, ICML). **The most important change is in eq. 5: debaters should keep the reward for their own side, or a bounded bonus for tool evidence the judge re-runs, because crediting both for moving the judge toward the decision pays one side to argue badly.**

Prior work and its limits

Label-free rewards.

- Self-rewarded training (SRT) rewards answers that match the majority of the model's own 16-64 samples (Shafayat et al. 2025). On Qwen2.5-Math-7B it first neared RL on true answers (0.32 against 0.33, trained on MATH-12K); on DAPO it declined after about 400-600 steps and collapsed by 1,200.
- Under GRPO (RL that scores each sample against others for the same question), random rewards raised Qwen2.5-Math-7B on MATH-500 by 21.4 points, against 29.1 for true rewards, but often did not help Llama3 or OLMo2 (Shao et al. 2026, ICML).
- Co-RL (Yang et al. 2026, preprint, August 2026) arranges models in a ring and rewards each for matching one peer's

majority vote. It gains 3.0-8.6% on average over 7 text benchmarks, with no chance correction and no protection against coordinated answers.
- For Task 3: the main experiment needs a random-reward control and a second model family. Co-RL is the closest baseline for the first-answer reward (eq. 4), which without its chance term and dependence weighting is close to Co-RL with many peers.

Judge rewards get hacked.

- Kenton et al. 2026 (preprint) train a generator and a critic by standard debate before a frozen, weaker judge on AIME-like math. Peak accuracy was 0.7474, against 0.7263 for RLAIF (RL on an AI judge's approval) and 0.7730 for RLVR (RL on automatically graded answers). The RLAIF model learned to impersonate the judge; critic hacking and game balance are among the stated limits.
- Self-play DPO (preference training) on the judge's probabilities, with quote validation, raised judge accuracy about 4 points, a "small but significant" effect (Arnesen et al. 2024, preprint).
- For Task 3: the debate-win baseline needs soft judge rewards, quote validation, word limits and a frozen weaker judge.

The PI's process rewards.

- FG-PRM (Li et al. 2025, Findings of EMNLP) is six step-level classifiers, one per hallucination type, trained on synthetic errors written from gold solutions. The paper uses them to detect errors and rerank sampled solutions; the parts read show no RL training. Choosing among 64 samples, it scored 0.94 on GSM8K and 0.57 on MATH against Math-Shepherd's 0.91 and 0.54 (LongFormer backbone), but its average detection F1 (combining precision and recall), 0.499 on synthetic and 0.484 on human-annotated data, is below ChatGPT's 0.513 and 0.531.
- HiPRAG (Wu et al. 2026, ICLR) is an RL process reward on each search decision, added to outcome and format rewards. It averages 65.4% (3B) and 67.2% (7B) on seven QA benchmarks and cuts unnecessary searches from over 27% in earlier baselines to 2.3%.
- For Task 3: credit moves from one agent's steps to many agents' messages, the label from gold solutions (FG-PRM's stated limit) to the group decision, and the step score from a trained classifier to the change in the judge's belief.

Why this method: the math behind it

Result	What it gives this task	Assumptions, and where it fails	Source
Uniform error rate η among K answers scales the clean risk by 1 - Kη/(K-1), plus a constant (Thm 1)	P6(i) for K answers	Symmetric loss and errors; $\eta <$ (K-1)/K	Ghosh et al. 2017, AAAI
Peer loss subtracts the loss on a random pairing; error rates $\rho 0$ and $\rho 1$ scale its expectation by 1 - $\rho 0$ - $\rho 1$ (Lemma 2); weight α^* keeps the optimum (Thm 4)	Bias cancellation in P6(ii); α^* for unbalanced topics	Yes/no; errors independent of the question given the truth; $\rho 0 + \rho 1 < 1$	Liu and Guo 2020
Agreement minus an other-task statistic pays constant or random reports 0; a trusted agent makes truthful effort "essentially the only equilibrium"	Structure of eq. 4; trusted anchors	Tasks alike in advance; coordination on cheap signals still pays	Dasgupta and Ghosh 2013; Shnayder et al. 2016; Gao et al. 2016, preprint
Dynamic potential-based shaping preserves optimal policies and equilibria	Eq. 5 for the group	Target fixed in advance; potentials logged when visited; every agent shaped on every transition	Devlin and Kudenko 2012, AAMAS

For P6, let the reference (the answer the Task 1.1 posterior favors without the agent's own group) err at rate η on every question, evenly over the K - 1 wrong answers and independently of the agent given the true answer Y. With the chance term at full weight, scoring the same answer a against the reference on a random hidden question from topic T, eq. 4's expected reward without confidence weights is (researcher's derivation):

$$\mathbb{E}[\, r \mid a, q \,] = \left(1 - \frac{K\eta}{K-1}\right)\Big[P(Y = a \mid q) - P_T(Y = a)\Big]$$

Here P_T(Y = a) is answer a's base rate in the topic, and dropping it leaves plain agreement up to a constant. So P6(i)'s 1 - 2η holds only for yes/no answers with symmetric errors, a reference independent of the agent given the truth, and η < 1/2; with K answers the factor is 1 - Kη/(K-1) (Ghosh et al. 2017), and with asymmetric errors plain agreement inherits the reference's bias (Natarajan et al. 2013, NIPS).

The chance term pays zero on average to answers that ignore the question, including a fixed answer colluders agree on (Dasgupta and Ghosh 2013; Shnayder et al. 2016). It also cancels the reference's bias, leaving the factor 1 - ρ0 - ρ1 for false-yes rate ρ0 and false-no rate ρ1 (Liu and Guo 2020, Lemma 2). Under symmetric errors plain agreement is already accuracy-optimal, so the chance term buys protection against collapse and collusion, and four problems remain:

- **Base rates.** At full weight the best answer is the one the evidence lifts most above its base rate, the most likely answer only on balanced topics. Liu and Guo's weight α* (Theorem 4) restores accuracy, but any weight below 1 pays a constant answer; batches balanced on the posterior answer within each topic, at full weight, get both.
- **Crediting.** As worded, the term "does not depend on the current answer", so under GRPO it cancels exactly when shared across a question's samples. It must be credited to the answer it scores.
- **Cheap signals.** Colluders who coordinate on cheap features of a question, such as wording or length, and also shape the reference still earn reward; Gao, Wright and Leyton-Brown 2016 show this is unavoidable when such signals exist.
- **All-ignore outcome.** If every agent ignores the question, all rewards are zero and learning stops, so trusted anchors are needed: tool-backed decisions and human audits (Dasgupta and Ghosh 2013, contribution iii).

For eq. 5, let Φ(h) be the judge's probability of the target decision after transcript h, logged as each message is read. Each message earns the change it causes, so the credits telescope:

$$\sum_{t=1}^{n} \left[\Phi(h_t) - \Phi(h_{t-1})\right] = \Phi(h_n) - \Phi(h_0)$$

For the group this is potential-based shaping (rewarding the change in a state's score, which keeps the best policy) of one end-of-debate reward, if the target is fixed before this debate's ruling (Devlin and Kudenko 2012). Each agent gets only its own credits, the group's gain minus the others', so per-agent invariance fails and the draft is right that policy invariance does not carry over. If the judge's probabilities are calibrated forecasts of its own final view, each credit is that message's advantage, the process reward that Setlur et al. 2024 and Jia, Rakhlin and Xie 2025 (preprints) justify.

Closest work and how this task differs

Work	What it does	What it assumes	What this task adds
Co-RL, Yang et al. 2026	Each model in a ring matches one peer's majority vote	Model diversity prevents shared errors	Left-out, dependence-weighted reference; chance term; message credit; P6
Kenton et al. 2026	Generator-critic standard debate RL against RLAIF, frozen weaker judge	Word limits balance the game	Target is not this judge's ruling; first answers rewarded without debate; audits
Qiu et al. 2026, preprint	Peer-prediction log-score with a subtracted marginal; DPO training	Large, diverse pools; collusion not considered	Agreement with a decision; audits; dependence weighting
CoMAS, Xue et al. 2026, ICLR	LLM judge's score split zero-sum between solver and evaluator; no labels	Judge scores track quality; no analysis of shared errors or collusion	Shared mistakes counted once; chance term; audits
MACA, Samanta et al. 2025, preprint	RL contrasting majority and minority reasoning traces "without external supervision"	Not stated in the abstract read	Left-out, dependence-weighted reference; chance term; P6
ACC-Collab, Estornell et al. 2025, ICLR	Turn credit: chance a discussion reaches the correct answer	Ground-truth labels	The same credit form without labels

The closest 2026 papers are Co-RL and CoMAS; a 2026 preprint also shows that a judge panel's mean keeps shared error that scores alone cannot identify, with a certificate built from two trusted anchors (Evaluator Ensembles Under Reward Hacking). The safest true claim is the combination, which the searches did not find: a chance-corrected reward against a left-out, dependence-weighted reference with guarantees (P6), dropped flagged agents, and label-free credit for each message. The web-search budget ran out before the August-October 2026 sweep was complete, so the novelty sweep should be re-run before submission.

What to borrow

Technique	Source	How to adapt it here
α^*-weighted chance term	Liu and Guo 2020, Thm 4	Estimate $\rho 0$ and $\rho 1$ from Task 1.1's confusion matrices, or balance topics
Spot-checks against ground truth	Gao et al. 2016	Use human-audited real decisions as an occasional training reward
Potentials logged at visit time	Devlin and Kudenko 2012	Store the judge's probability as each message is read; never recompute with a later judge
Counterfactual credit	Foerster et al. 2017, COMA	Re-query the judge without the message, or with a resampled one
Collapse controls	Shafayat et al. 2025; Zhang et al. 2026, Co-rewarding, ICLR	Early stopping on audited decisions; slowly updated reference agents

Fixes to the white paper

- **Section 3 claims.** Kenton et al. 2026 is summarized misleadingly: its main finding is that debate training reduces reward hacking relative to RLAIF. The debate critic fabricated quotes only before a deliberately weakened judge (chain of thought off, fewer than 50 visible words), and the approval-trained model impersonated the judge (about 90% of samples by 30% of training). Shafayat et al. 2025 is fair but should say the majority is the model's own samples, and that accuracy first neared ground-truth training (0.32 against 0.33), then collapsed to one template answer, which is optimal for that reward. Wen et al. 2025 is fair with "human approval" and "time-limited evaluators" (3-10 minutes; false positives up 24.1% on QuALITY and 18.3% on APPS; ICLR 2025 venue unverified).
- **Suggested Section 3 wording.** "Nor can agents yet keep improving without an answer key. Rewarding agreement with a model's own majority vote helped at first, then collapsed accuracy as the model learned to give one answer to every question (Shafayat et al., 2025). Rewarding an AI judge's approval taught a model to impersonate the judge, and even debate training, which curbed this, let a critic fabricate quotes before a weakened judge (Kenton et al., 2026). Rewarding human approval taught models to convince time-limited evaluators that wrong answers were right (Wen et al., 2025). Newer methods that reward agreement among several models report stable training through model diversity, but offer no guarantee against shared or coordinated answers (Yang et al., 2026)."
- **P6.** Restate it with the assumptions and four problems above, adding for the panel: "Subtracting how often the same answer matches the reference on an unrelated question from the same topic pays, on average, nothing to any answer that ignores the question, including a fixed answer colluders agree on in advance."
- **Eq. 5 and the Task 3 text.** "Credits each message for helping reach the decision rather than for winning" holds only if the target comes from outside this debate's ruling, such as the Task 1.1 posterior plus re-run tool evidence. Crediting both debaters toward the decision pays the one assigned to the other answer to argue badly, the collusion Task 2.2 flags. Suggested: "Debaters keep the standard zero-sum reward, the judge's final probability of their assigned side (Arnesen et al., 2024), or earn only a bounded bonus for tool evidence the judge re-runs; the decision reward trains the judge and first answers." Compare both options in the main experiment.
- **Main experiment and citations.** Add Co-RL, peer prediction (Qiu et al. 2026), random rewards (Shao et al. 2026), an RLVR ceiling and a second model family. Cite Co-RL as Yang et al. 2026 (arXiv 2608.17253), and Lennie et al. 2026 as a LessWrong post by "lennie, joanv, Shi and J. Pfau" (MATS 9.0, July 2, 2026). If the draft cites Casheekar and Tangirala 2026 (preprint), say that it trains PPO agents on witness graphs and is not a study of LLMs. Describe FG-PRM as detection and reranking, and do not claim its detection F1 beats ChatGPT's.

Draft related-work paragraph

> Rewarding agreement with a model's own majority vote helped at first, until the model gave one answer to every question (Shafayat et al., 2025). Peer majorities (Yang et al., 2026) and AI judges' scores (Xue et al., 2026) also train agents without labels, but neither work analyzes collusion. Rewarding a judge's approval taught a model to impersonate the judge; standard debate curbed this but let a critic fabricate quotes before a weakened judge (Kenton et al., 2026). We reward each first answer for matching a reference built without the agent's own group. We subtract the same answer's agreement with the reference on another question from the topic (Dasgupta and Ghosh, 2013; Liu and Guo, 2020), so answers that ignore the question earn nothing on average. Unlike earlier turn-level credit in discussion, which needs the correct answer (Estornell et al., 2025), we credit each message by how far it moves the judge toward a decision fixed outside the debate. Debaters earn reward only for their own side or for tool evidence the judge re-runs. This extends our process reward models (Li et al., 2025; Wu et al., 2026) to many agents' messages.

Evaluation: testbeds, red team and baselines

Each testbed can start from public data with known answers; the maritime scenarios and planted debates (debates on known-answer questions) must be built. MEQA as published builds each question from one document (Li et al. 2024), so multi-document maritime questions extend it. **The most important change is a red team that chooses when to attack, at four levels of attacker power (ACL-1 to ACL-4 in Korbak et al. 2025, preprint).** Such attack selection cut safety, the chance that no attack gets through, from 99% to 59% at a 0.5% audit budget (Schaeffer et al. 2026, preprint).

Testbed A: math and code with a shared flawed hint

Testbed A borrows a protocol that poses each question with and without a hint and keeps pairs that switch to the hinted answer (Chen et al. 2025, preprint). Giving one flawed hint to k of N agents makes the group sharing a mistake known by construction, and each agent's switch rate shows how easily it is misled.

When a hint changed an answer, the reasoning mentioned it in only 25% (Claude 3.7 Sonnet) and 39% (DeepSeek R1) of cases on average (Chen et al.). Shared errors must therefore be inferred from who agrees with whom, as in Task 1.1.

Dataset or protocol	What it gives	Size	Access	How we would use it
Paired hints (Chen et al. 2025; Turpin et al. 2023)	Six hint types (Chen); accuracy drops of up to 36% (Turpin)	MMLU, GPQA; 13 BIG-Bench Hard tasks	Preprints; public data	One hint for k of N agents; vary hint type and k
ProcessBench (Zheng et al. 2024)	Earliest wrong step, marked by doctoral-level annotators	3,400 solutions	Preprint; HuggingFace; no license stated	Disputed-step labels (P4); 18.8–51.8% of correct-answer solutions in harder subsets hide a wrong step
CodeCrash; ImpossibleBench	Misleading hints and structural changes (23.2% average drop); specifications contradicting unit tests	1,279 questions; LiveCodeBench, SWE-bench tasks	NeurIPS 2025; ICLR 2026	Shared hint for code; cheating incentive, unit tests as automatic audits
Password-locked models; sandbagging	Skill hidden unless a password appears; prompted underperformance	n/a	Preprints	Agents of set skill, re-measured on held-out items
MATH; GSM8K; LiveCodeBench	Problems with answers or unit tests	12,500; 8.5K; 400	Public	Source items; LiveCodeBench's date split limits contamination

Testbed B: multi-document questions and maritime scenarios

Testbed B asks questions that need several documents, such as whether a vessel has left port, and plants outdated or forged reports among them. Its closest competitor is MADAM-RAG, in which agents discuss conflicting documents and an aggregator combines their answers (Wang et al. 2025, venue not confirmed).

AIS, the position and status reports that ships broadcast, may be spoofed (MARAD advisory 2026-015), and 72–76% of industrial fishing vessels are not publicly tracked (Paolo et al. 2024, Nature). Real AIS tracks should therefore only seed scenarios, and the scenario generator must define each true answer.

Dataset or protocol	What it gives	Size	Access	How we would use it
MEQA	Multi-hop event questions (chaining facts) with explanations, one document each	2,243 questions, 211 documents	GitHub, MIT license	Base pipeline, extended to several documents
RAMDocs, MADAM-RAG	Ambiguous queries, misinformation, noisy documents	Not in abstract	Preprint	Required baseline; gains up to 15.80% (absolute) on FaithEval
ConflictBank; ConfRAG	Misinformation, temporal and semantic conflicts; real contradictions in web text	553,117 question-answer pairs; 1,814 questions	Preprint; ACL 2026	Outdated-schedule templates; held-out real conflicts
MarineCadastre; Danish AIS	Positions, speeds and status codes	US waters from 2009; Danish waters	CC0 (public domain) per repository; Danish terms not stated	Seed tracks; departure as status 5 (moored) to 0 (under way), after confirming code ranges
Global Fishing Watch	Port visits: entry within 3 km of an anchorage, exit beyond 4 km	Global	API token; non-commercial only	"Left port" labels; low-confidence visits may be false
Paolo et al. 2024; xView3-SAR	Satellite vessel detections matched to AIS	2017–2021; nearly 1,000 radar scenes	"Freely available"; registration; no license seen	Public stand-in for the satellite-image audit

Testbed C: scientific claims and peer review

Testbed C has agents judge scientific claims and papers, with replication outcomes as human audits. The best human forecasts of these outcomes were right 71–78% of the time (Dreber et al. 2015; Gordon et al. 2021; SCORE), while SCORE's machine forecasts "were not consistently effective".

Older outcomes, such as the 2015 Open Science Collaboration replications (summary), may be in training data, so 2026 releases with anonymized claims make safer audits. The labels are also noisy: in Many Labs 2, effects varied significantly across samples for 11 of 28 findings (Klein et al. 2018).

Dataset or protocol	What it gives	Size	Access	How we would use it
SCORE; Institute for Replication (both 2026)	Replication, reproduction and reanalysis outcomes; SCORE adds forecasts	49% of 164 papers replicated; 110 papers	OSF, Zenodo; formats and licenses not checked	Fresh human-audit labels
Gordon et al. 2021	Replication outcomes with prediction-market (betting) and survey forecasts	103 replications	PLOS ONE; R package pooledmaRket	Human collective baseline; markets 73% accurate
Schroter et al. 2008; SPECS (AAAI-26 pilot, preprint)	Major errors planted in papers; LLM-perturbed papers with a detection score	9 errors in each of 3 papers; 783 papers	JRSM; SPECS release not confirmed	Planted cases; reviewers found 2.58 of 9 errors, AI detection rose from 0.43 to 0.64
SPOT	Real errors that led to errata or retractions	83 papers, 91 errors	Preprint	Held-out real errors; the best model found at most 21.1%
PeerRead; NLPeer	Reviews; PeerRead adds accept or reject decisions	14.7K drafts, 10.7K reviews; over 5,000 papers, 11,000 reviews	NAACL 2018; preprint; licenses not checked	Disagreement among human reviewers

Dataset or protocol	What it gives	Size	Access	How we would use it
Agents4Science	Reviews by three LLMs per paper; human scores for 79 papers	253 reviewed papers	Preprint; reviews public	Correlated AI reviewers (Pearson 0.48)

Red team

AI control evaluations let a red team attack a fixed protocol with a small audit budget, 2% of solutions in Greenblatt et al. 2024 (ICML). Of the five levels in Korbak et al. 2025, ACL-5 is out of scope: its authors say "we cannot determine rules for a control evaluation".

The red team must include attack selection, recognizing planted debates, sandbagging (arguing badly on purpose), prompt injection (instructions hidden in text the judge reads), true but selectively chosen facts, steganography (hidden signals between colluders) and attacks spread across agents. In the table, only Hu et al. (ACL 2026) has a confirmed venue.

Source	What to borrow	How it maps to levels 1–4
Korbak et al. 2025	Red-team powers scaled to model capability	1: prompts, poisoned documents; 2: fine-tuned covert encoding; 3: goals held in weights, copies coordinating; 4: any trained attack
Schaeffer et al. 2026	Attack selection as classification; safety hinges more on the monitor's false-positive rate than its true-positive rate	1–4: cheat only when no audit is expected; their attacker was only prompted
Terekhov et al. 2025; Hu et al. 2026	Prompt injection hidden in outputs; collusion with true evidence fragments (70.6–74.4% attack success)	1–2: attacks on the judge that re-running evidence does not catch
Taylor et al. 2025; van der Weij et al. 2024	Selective sandbagging; a blue team could not reliably spot it (Taylor)	2–4: debaters or judges that argue badly on purpose
Needham et al. 2025; Hubinger et al. 2024	Telling evaluations from real use (AUC 0.83 for Gemini-2.5-Pro, 0.92 for humans; 0.5 is chance); backdoors that survive safety training	2–3: behave well whenever a debate looks planted; report planted-versus-real AUC
Kitaoka et al. 2026; Pang et al. 2026	Attacks spread across agents (7% flagged, against 87% for one agent); covert channels needing no shared key	4: attacks spread across debates; audits must look beyond transcripts

Baselines and metrics

Majority vote at equal compute is the bar to beat, since voting explains most gains credited to discussion (Choi et al. 2025, NeurIPS). For categorical answers, averaging with equal weights is essentially majority vote; keep it only for numeric scores, and add the rater model and multi-family panel below. For free-form answers, add universal self-consistency, where an LLM picks the most consistent sampled answer (Chen et al. 2023, preprint).

Method or metric	Who uses it	Why include it
Best single judge	Kohli 2026, preprint	Matched or beat a nine-judge panel worth about two independent votes
Majority vote at equal compute	Wang et al. 2023; Choi et al. 2025	Choi et al. prove discussion alone does not raise expected correctness
Rater model with independent errors	White paper; Dawid and Skene 1979 (unverified)	Shows what counting shared mistakes once adds (P1)
Panel from different model families (PoLL)	Verga et al. 2024, preprint	Beat one large judge at over 7 times lower cost
Discussion; standard debate with a fixed judge	Du et al. 2024; Khan et al. 2024	What the project replaces; isolates impartial choice and the harness

Method or metric	Who uses it	Why include it
Effective number of judges (independent votes a panel is worth); error correlation; CAPA (chance-adjusted shared mistakes)	Hossain et al. 2026, preprint; Goel et al. 2025, venue not confirmed; Kim et al. 2025, ICML	Shared error per condition (P1, P2); held-out model families can still share errors
Calibration error (gap between confidence and accuracy); accuracy on the disputed step	Guo et al. 2017; ProcessBench	The debate threshold (eq. 3) needs calibrated posteriors; P4 concerns single steps
Wrong decisions that get through, usefulness (rescaled accuracy) and false flags at 0.5%, 2% and 5% audit budgets	Greenblatt et al. 2024; Schaeffer et al. 2026	Main red-team result; prior work used 0.5% and 2%

Fixes to the white paper

- **Testbed B.** Reconcile MEQA's question count: 2,243 in the paper, 2,093 in the README. Suggested: "multi-hop event questions from our MEQA pipeline, extended to several documents with outdated, forged or selectively chosen reports, including maritime scenarios seeded with public vessel tracks." No formal export-control review was done.
- **Testbed A.** "Agents of known skill" needs a mechanism, since imitating a weaker model had only "mediocre success" (van der Weij et al. 2024). Suggested: "agents whose skill we measure on held-out items; k of N agents receive the same flawed hint (Chen et al., 2025)."
- **Red team.** Training colluders is ACL-2 or higher, beyond Greenblatt et al. 2024, whose red team prompted and scaffolded GPT-4. Suggested: "a red team at four levels, from prompting to training arbitrary attack strategies (Korbak et al., 2025)."
- **Audit rate.** Griffin et al. 2024 (preprint) cast control evaluations as partially observable stochastic games (players see only part of the state) and reduce them to zero-sum games, so existing algorithms can find protocols with the best safety-usefulness trade-offs. Suggested, as the mathematical reasoning the program officer requested: "We set the audit rate and the share of planted debates by solving a game between the red team and our protocol (Griffin et al., 2024)."
- **Baselines.** Replace "averaging with equal weights" with the draft paragraph's list.
- **Testbed C.** Suggested: "replication outcomes released in 2026 serve as human audits, treated as noisy labels with an estimated error rate," matching the draft's P6.
- **Novelty.** The search found no study that tracks shared error as one hint reaches more models, and no control evaluation that protects a judge's decision in a debate. It was not exhaustive, and its 2025–2026 sweep was under-sampled because the search budget ran out, so both gaps are provisional. Suggested: "To our knowledge, no prior control evaluation protects decisions made by a judge in a debate."
- **Section 2, line 39.** The Amayuelas et al. 2024 (preprint) abstract gives no number for "sharply cut"; cite one from the full paper or soften it.

Draft evaluation paragraph

> All data are public or synthetic. (A) Agents whose skill we measure on held-out items solve math and code problems. One flawed hint goes to k of N agents (Chen et al., 2025), so we know who shares a mistake, and expert labels mark the first wrong step (Zheng et al., 2024). (B) We extend our MEQA pipeline (Li et al., 2024) to several documents with outdated, forged or selectively chosen reports. Maritime scenarios start from public vessel tracks, but our generator fixes the true answer. (C) For scientific claims and peer review, replication outcomes serve as human audits, and papers with planted errors (Schroter et al., 2008) as automatic audits. Following AI control (Greenblatt et al., 2024), a red team at four levels (Korbak et al., 2025) makes debaters and judges collude, recognize planted debates, and attack only when they expect no audit. We count wrong decisions that get through at audit budgets of 0.5% to 5%. Baselines are the best single judge, majority vote at equal compute, a rater model with independent errors, a panel from different model families (Verga et al., 2024), discussion (Du et al., 2024), and standard debate (Khan et al., 2024).